

Algorithmic Game Theory - AI802

Lecture 1 - Introduction

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Game Theory = Study of decision-making in settings with multiple rational* agents whose outcomes depend on each other's choices.

*Rational = choosing actions to best achieve one's objective, given the strategic situation.

**Algorithmic Game Theory = Studying Game
Theory with a Computational Lens**

Scenario that you faced?

Can you come up with one scenario where you felt, you were part of a “game”?

System/Mechanism Design

Video Discussion on the Badminton Match

What are the players trying to do? What are your thoughts?

Tournament Bracket

London 2012 Olympic Games – Women's Doubles (Badminton)

Tournament Format

1. GROUP STAGE (Round Robin)

16 pairs divided into 4 groups (A, B, C, D)
Top 2 from each group advance (8 pairs)

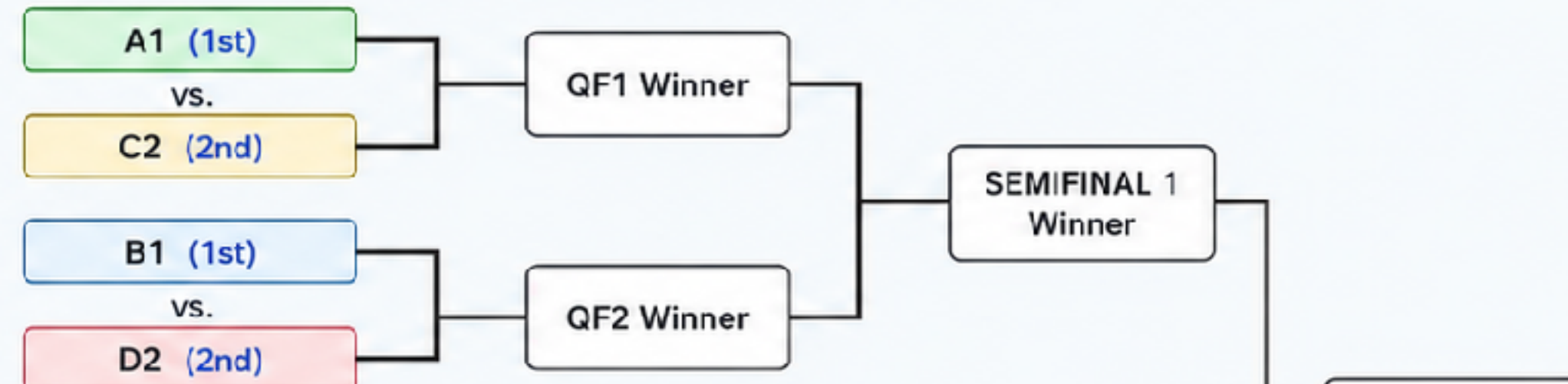
GROUP A	GROUP B	GROUP C	GROUP D
A1 (1st)	B1 (1st)	C1 (1st)	D1 (1st)
A2 (2nd)	B2 (2nd)	C2 (2nd)	D2 (2nd)
A3	B3	C3	D3
A4	B4	C4	D4

Top 2 from each group qualify
for Knockout Stage (8 pairs)

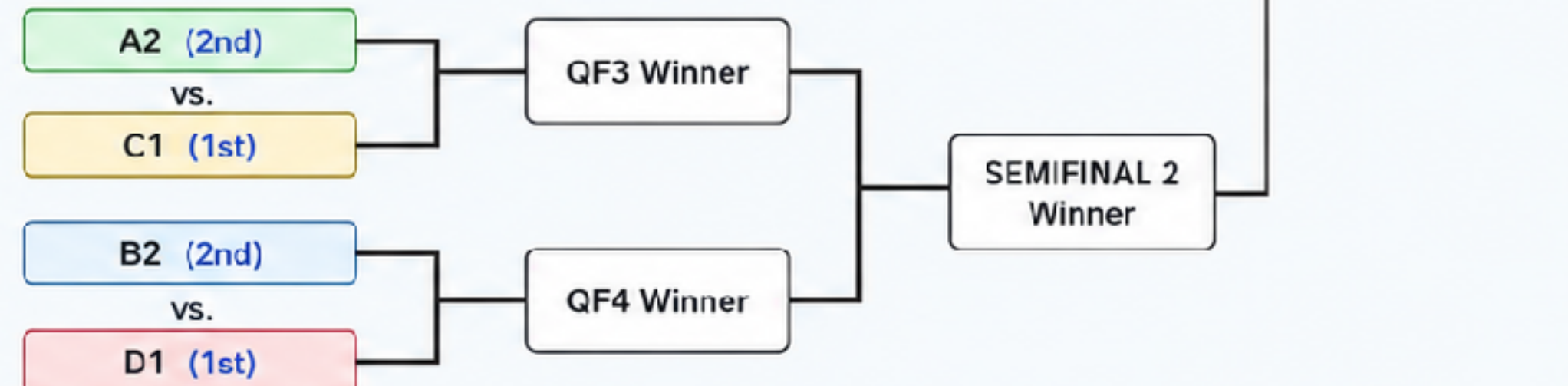
2. KNOCKOUT STAGE (Fixed Bracket)

Bracket is pre-determined by group finishing position.
Teams know their potential path before the final group match.

TOP HALF



BOTTOM HALF



- A1, B1, C1, D1 = Group Winners
- A2, B2, C2, D2 = Group Runners-up

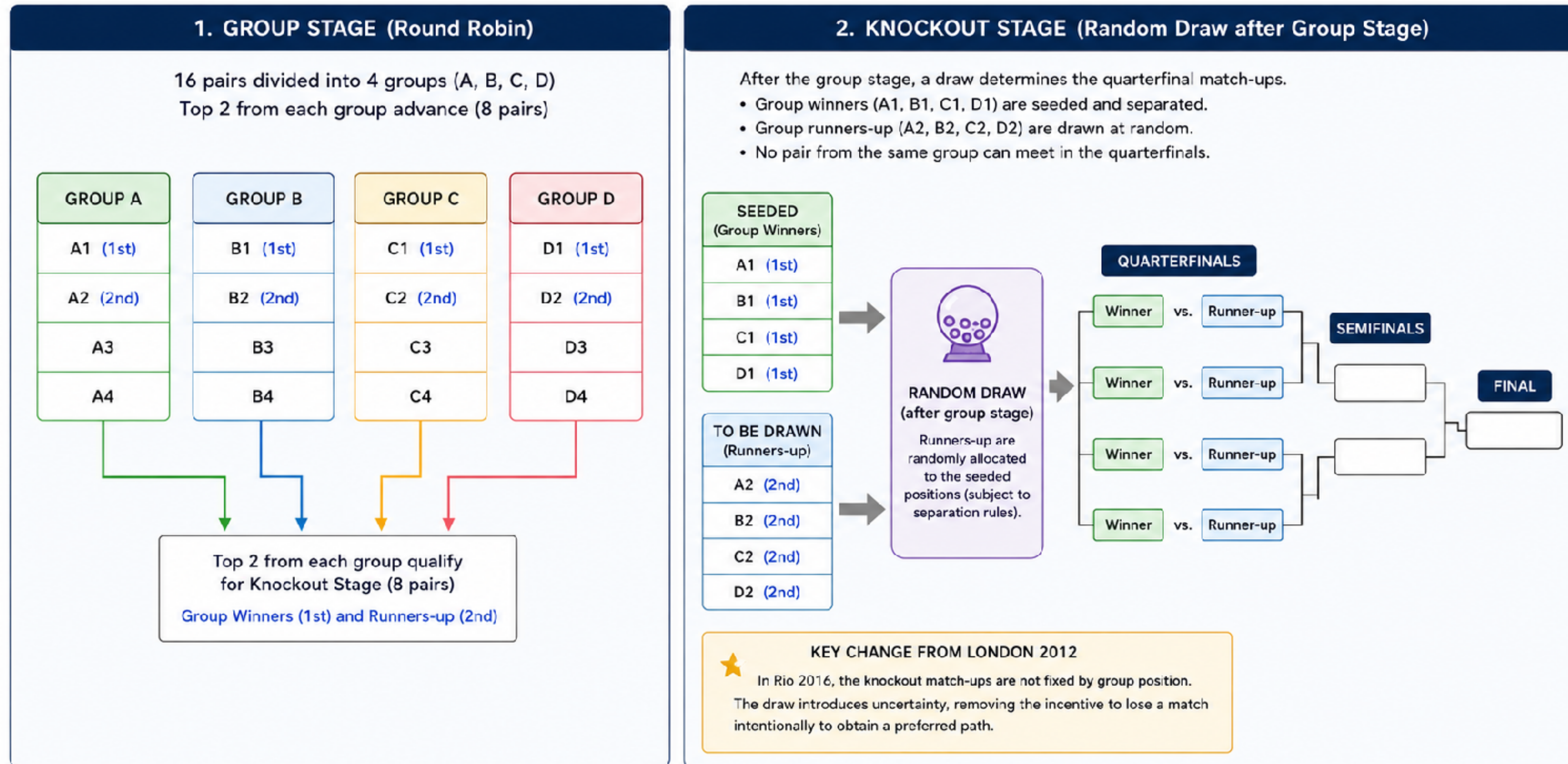
Video Discussion on the Badminton Match

Any suggestions on how the tournament format must be changed?

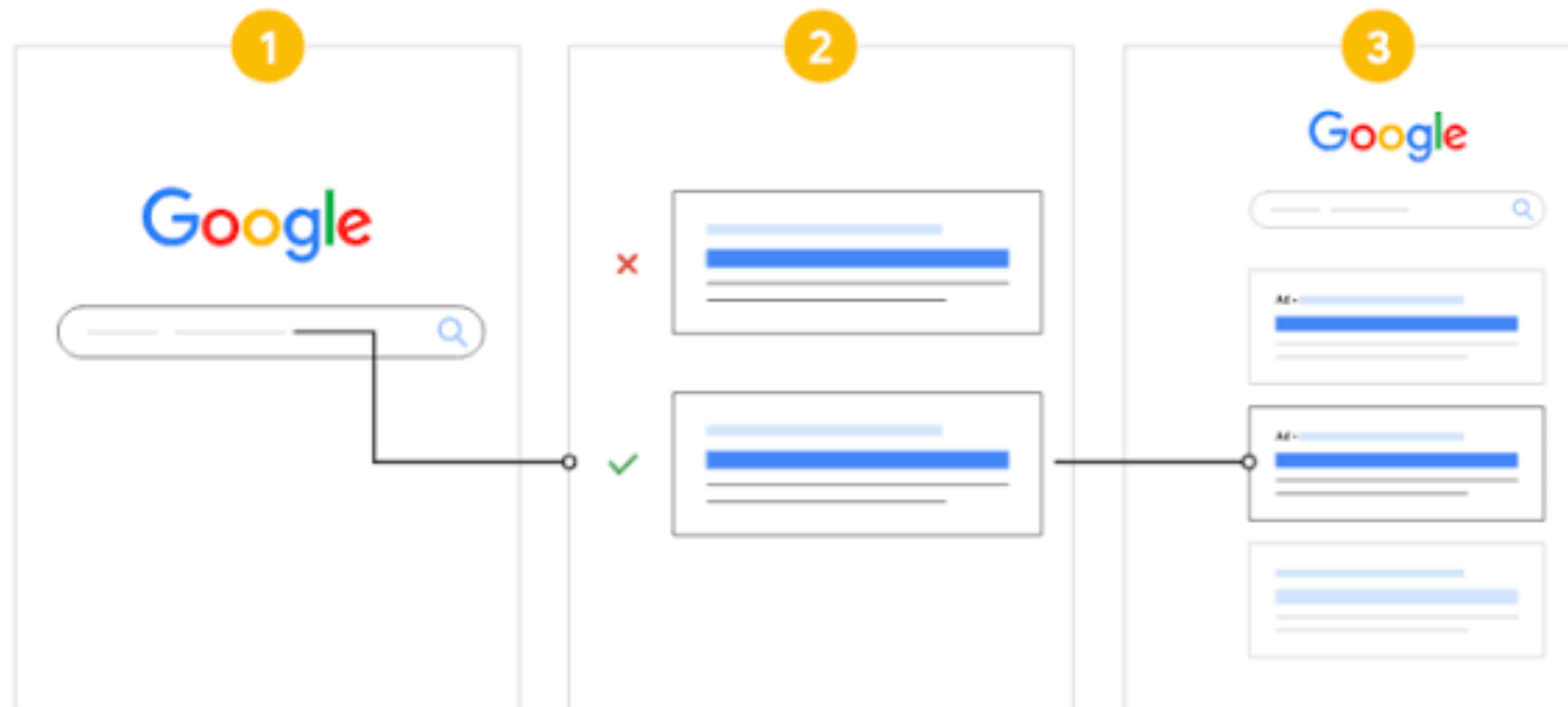
How it Changed for Rio Olympics

Rio 2016 Olympic Games – Women's Doubles (Badminton)

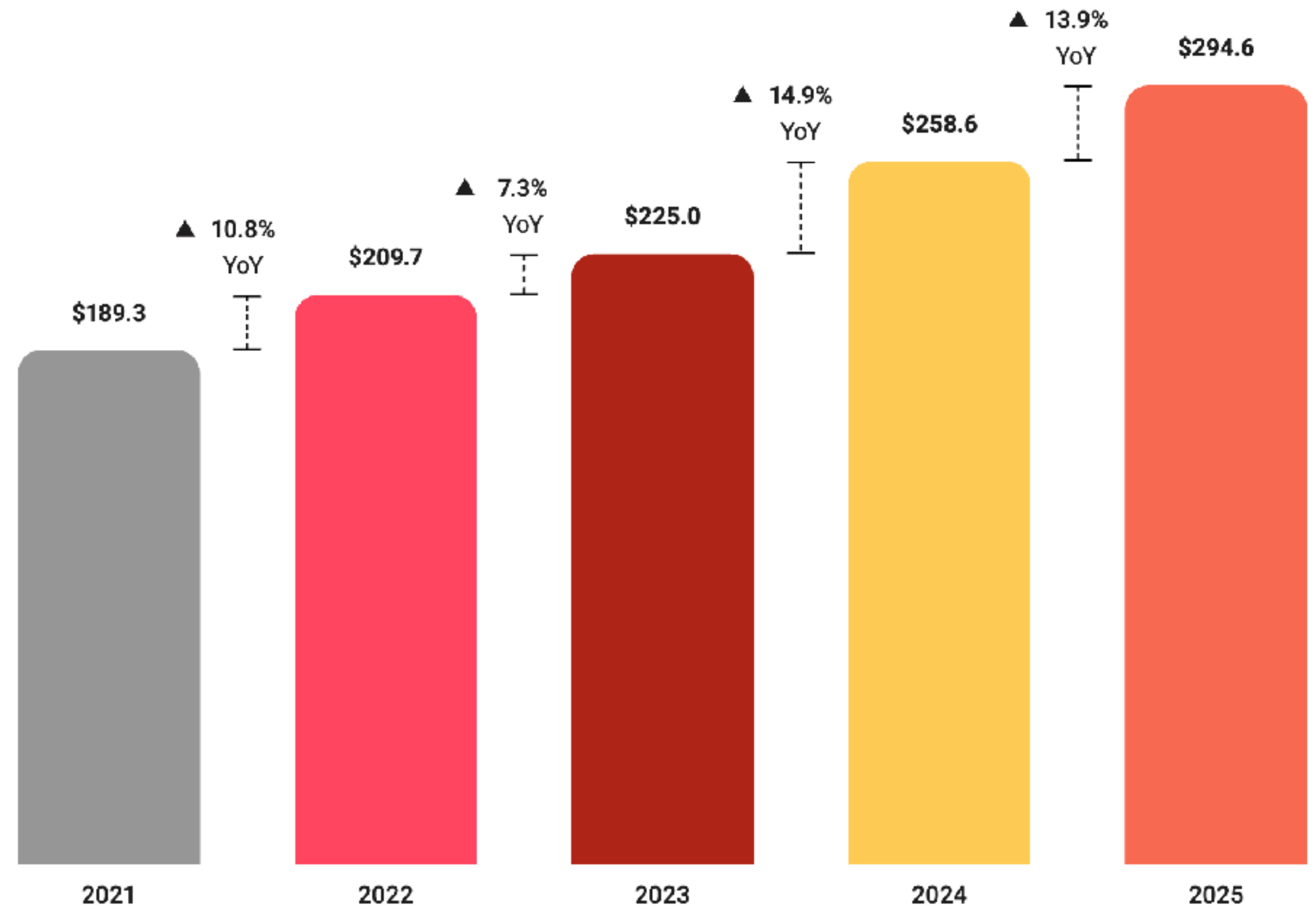
Tournament Format (Revised from London 2012)



Auctions

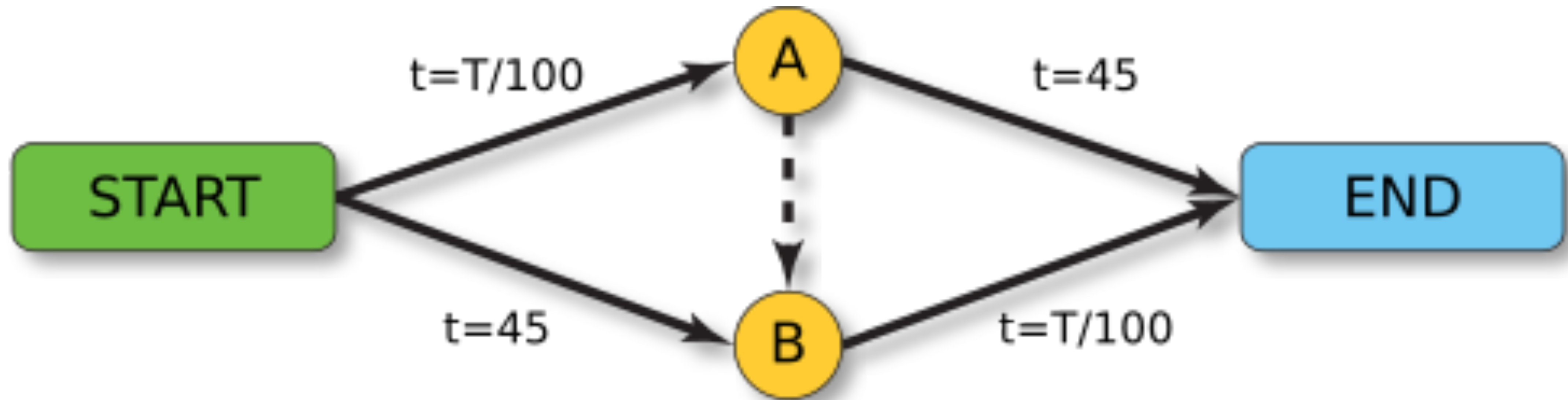


Five-year (2021-2025) internet advertising revenue trend (\$ billions)



Source: IAB / PwC Internet Ad Revenue Report, FY 2025.

Routing Games and designing roads



Opening shortcuts can make the traffic worse!

Cooperation and Incentives

If everyone benefits when countries reduce emissions, why might each country want to be “free-riders”?



AI for Games

AI for Multiplayer Games



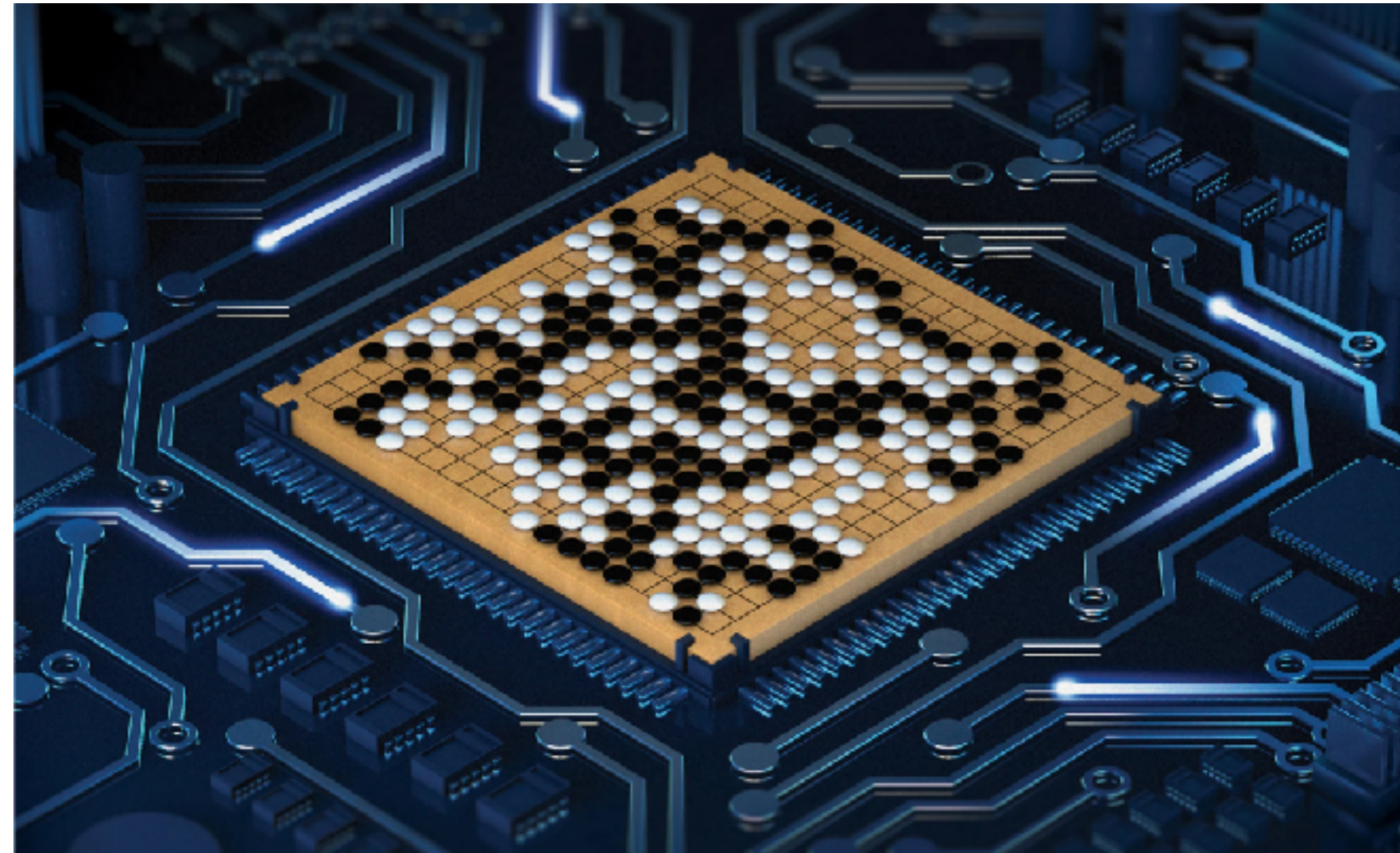
Pluribus Poker Bot



AlphaStar

Key Questions: How to handle very large action/state spaces and imperfect information?

AI for Single Player Game



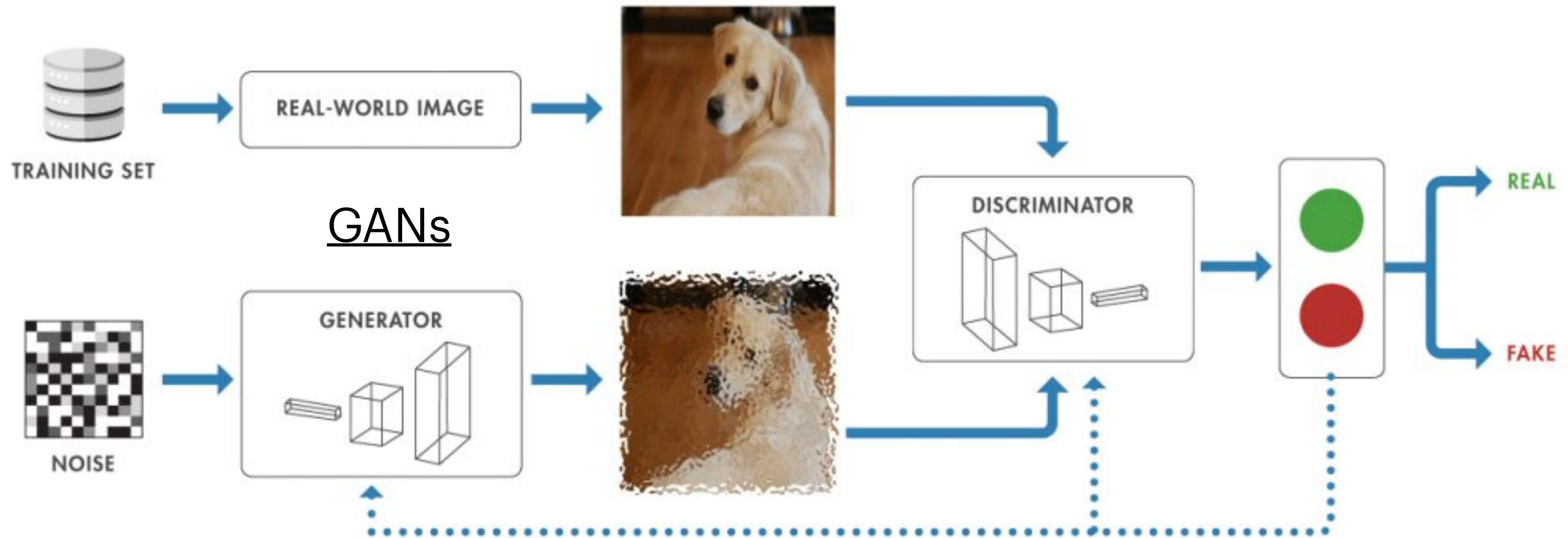
AlphaGo

Trained with “self-play”,
where the algorithm plays many
games against itself.

Key Question: Why does self-play seem to work?

Machine Learning, XAI (Explainable AI)

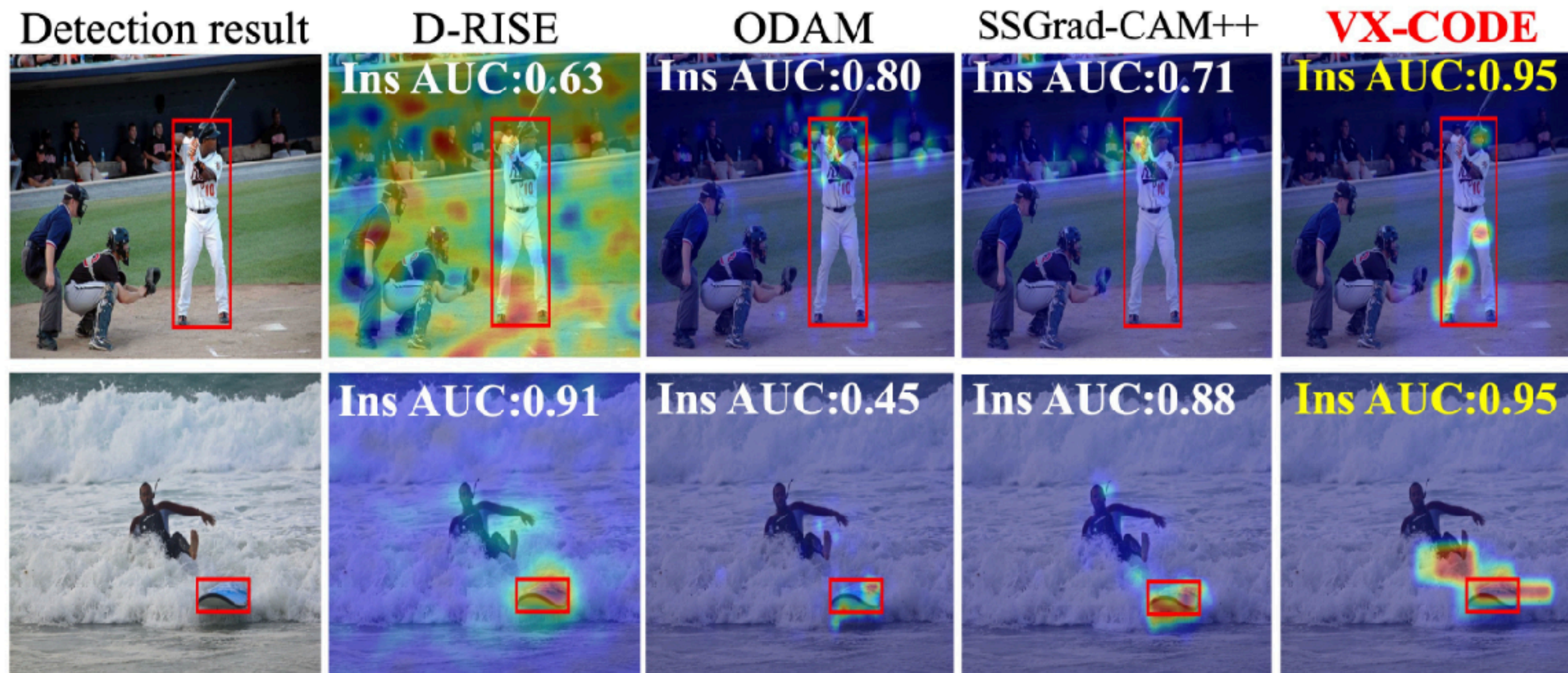
Image Generation with GANs



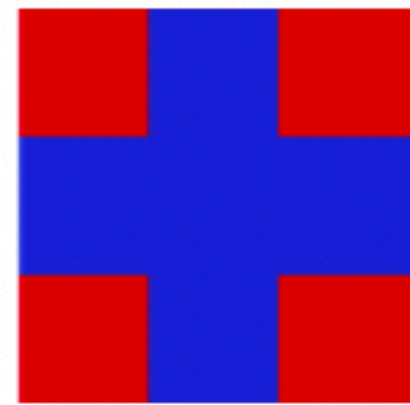
$$\min_{\theta_G} \max_{\theta_D} \mathbb{E}_{x \sim p_d(x)} [\log D(x; \theta_D)] + \mathbb{E}_{z \sim p_z(z)} [\log (1 - D(G(z; \theta_G); \theta_D))]$$

Key Question: How to solve the above problem?

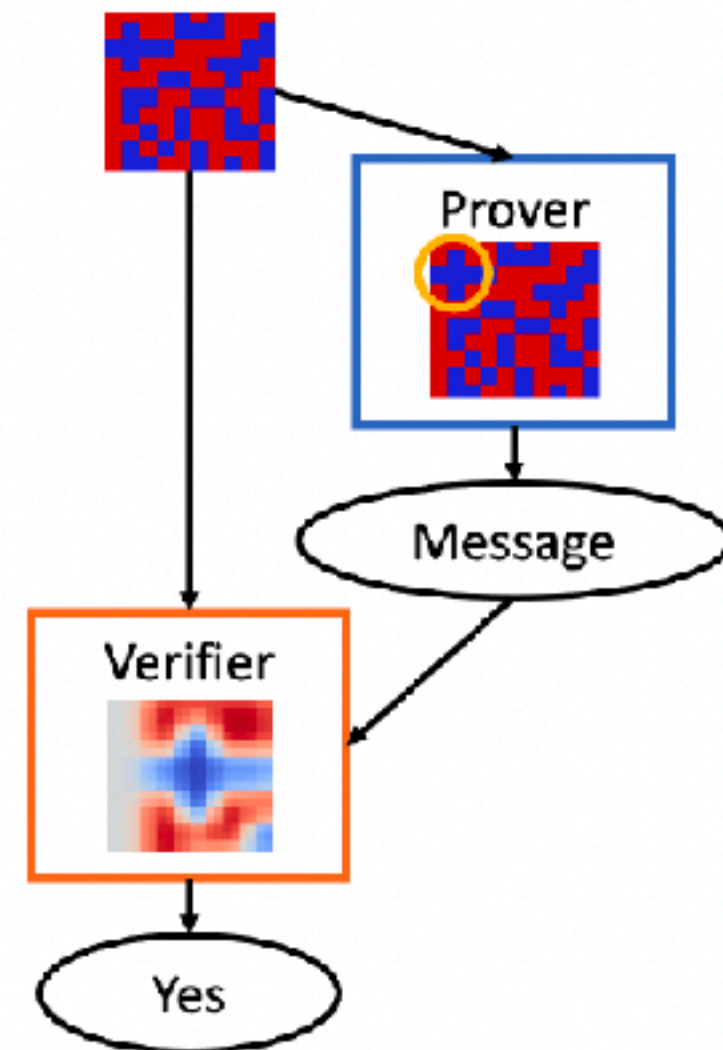
Can you explain a classifier's decisions?



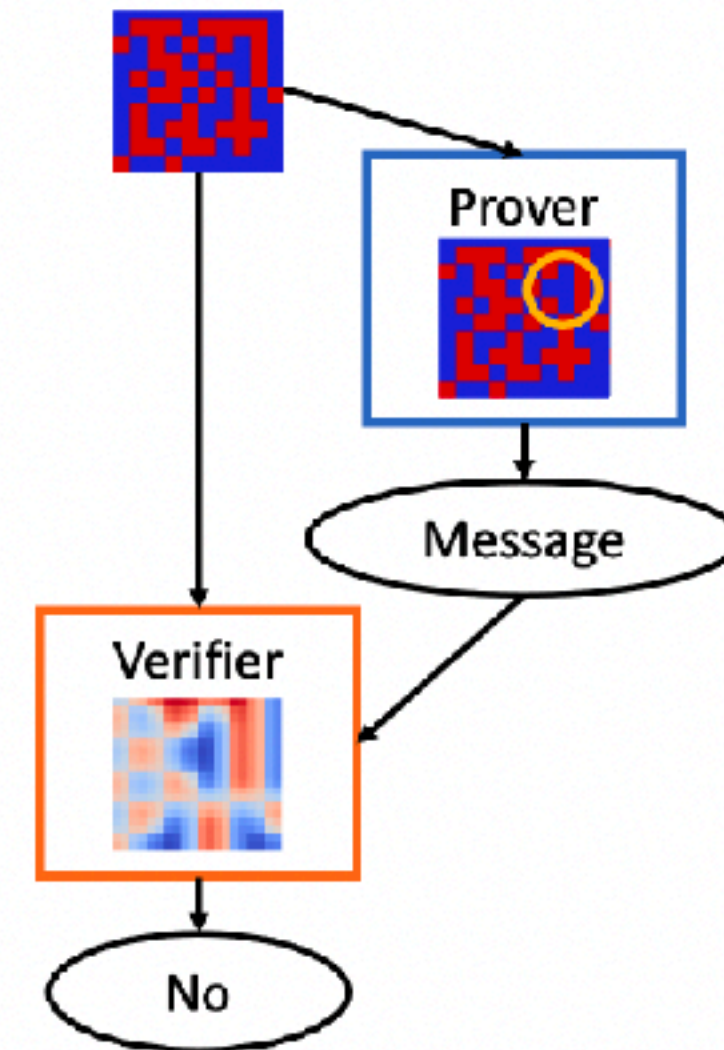
Prover-Verifier Games



(a) Pattern to detect.



(b) When the pattern exists.



(c) When the pattern doesn't exist.

Prover-Verifier Games

Key Question: How to build a *reliable* Verifier and Prover, to increase model explainability?

Can Debate help improve accuracy and Scalable Oversight?

Prompt: Give a bullet point biography of David S. Johnson highlighting their contributions and achievements as a computer scientist, with each fact separated with a new line character.

Round 1

Agent 1: -- He earned his bachelor's degree from **Dartmouth College** in 1967 and his Ph.D. from MIT in 1973. ❌

-- Johnson has received numerous awards for his research and contributions to computer science, including the **ACM Turing Award** in 2013. ❌

-- Johnson's research interests include optimization, algorithms, and complexity theory. ✅

Agent 2: -- David S. Johnson is an American computer scientist and mathematician. ✅

-- Johnson was elected to the National Academy of Engineering in 2006. ✅

-- He has been awarded numerous honors and awards for his contributions to computer science, including the **ACM SIGMOD Innovations Award** and the Knuth Prize. ❌

Round 2

Agent 1: -- David S. Johnson is a computer scientist and mathematician. ✅

-- Johnson was elected to the National Academy of Engineering in 2006. ✅

-- He has received numerous awards for his contributions to computer science, including the **ACM Turing Award** in 2013 and the Knuth Prize. ❌

Agent 2: -- David S. Johnson is an American computer scientist and mathematician. ✅

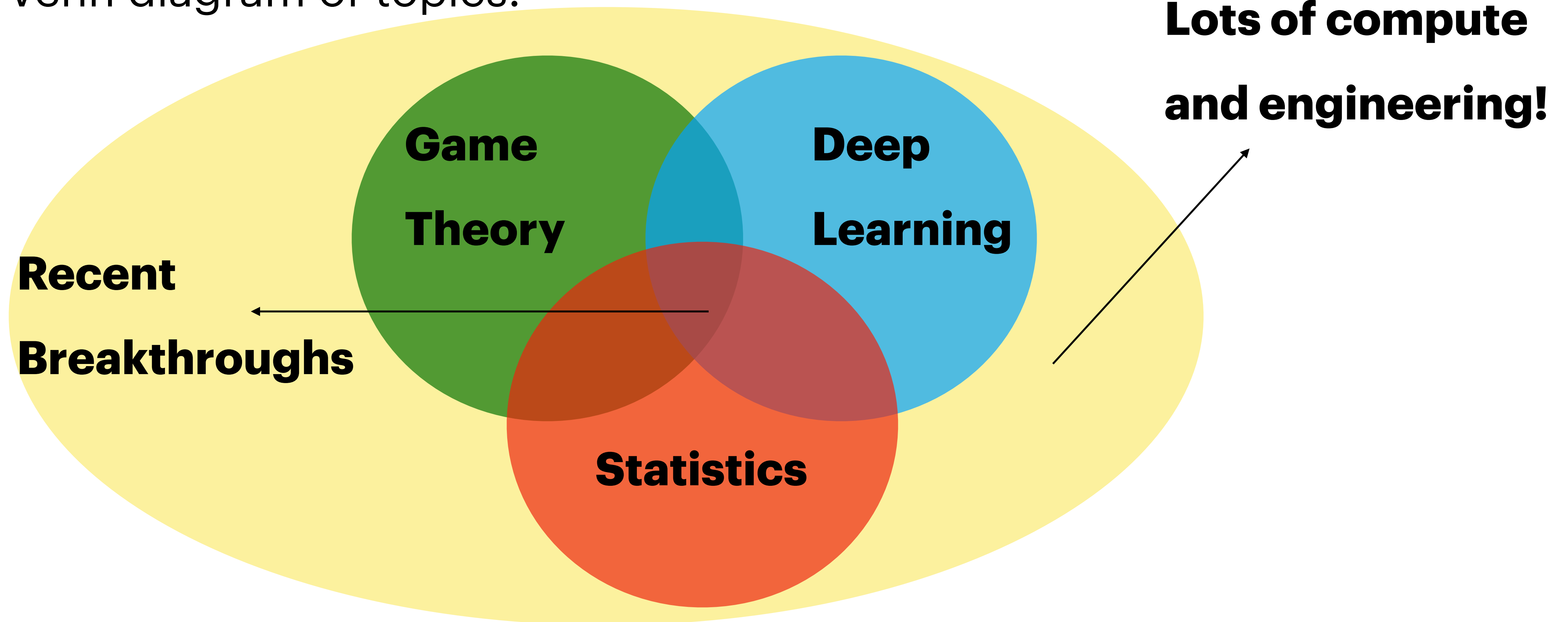
-- He has worked at several prestigious universities, including Columbia University, AT&T Bell Laboratories, and **Cornell University**. ❌

-- Johnson is known for his work in the field of algorithms, particularly in the areas of network optimization, combinatorial optimization, and scheduling theory. ✅

Debate

What's Common?

- A Venn diagram of topics:



What we will cover

- Auctions
- Selfish Routing
- Bimatrix games and equilibria.
- Decentralised learning of equilibria.
- Introduction to minmax optimization and GANs.
- Cooperative Games and Shapley Values.
- Extensive form games and analysing (Kuhn) Poker.

What is not covered

- We will not go too much into computational complexity (we will be aware of it).
- Multiagent Reinforcement Learning.
- Deep Learning and related training methods.

Learning Outcomes

- Mathematically formulate any (relevant) problem as a multagent learning problem.
- Analyse effectiveness of systems/mechanisms.
- In real-world strategic scenarios, identify the players and the game.
- Identify the information available and use a relevant algorithm.
- Identify relevant equilibrium concepts/notions and be aware of its complexity.
- Peek behind the recent AI breakthroughs and abstract out the relevant mathematical concepts.

Relevant Material

- **Game Theory Basics:** Lecture Notes on Algorithmic Game Theory by Tim Roughgarden, Algorithm Design and Analysis by Noam Nisan, Tim Roughgarden, Éva Tardos.
- **Misc:** Relevant papers/readings will be posted for certain weeks.

Schedule (Subject to some changes)

Week	Topic
36	Introduction and Recap
37	Auctions
38	Auctions
39	Selfish Routing
40	Selfish Routing
41	Bimatrix Games and Equilibria
42	Break
43	Decentralised Learning of Equilibria
44	Minmax Optimization and GANs

Schedule (Subject to some changes)

Week	Topic
45	Extensive Form Games and Poker
46	Cooperative Games , Shapley Values and Explainability
47	Recap of certain topics and exam related discussions

Special Concession and In-Class Expectation

- Since we have very few students, there is a risk of 1/no student attending.
- Instead I request you to cooperate with each other and ask me to reschedule **up to three** classes. Choose wisely!
- Our last teaching week is Week 49.
- Please consider this is a reading group, since it is a very small class.
- Let us have this policy that there are no bad ideas, when we are thinking about a new problem! So please feel free to talk about the material, questions and assumptions, provide suggestions for how to proceed etc.